

EX1 $X \sim N(7, 2)$

$$P(X < 9) = P\left(Z < \frac{9-7}{\sqrt{2}}\right) = \Phi(1) = 0,84134$$

$$P(X > 8) = 1 - P(X \leq 8) = 1 - P\left(Z < \frac{8-7}{\sqrt{2}}\right) = 1 - \Phi(0,5) = 1 - 0,69146 = 0,30854$$

$$P(4 < X < 11) = P\left(\frac{4-7}{\sqrt{2}} < Z < \frac{11-7}{\sqrt{2}}\right) = P(-1,5 < Z < 2) = \Phi(2) - \Phi(-1,5) = 0,97725 - 0,06681$$

~~$$P(X < 11) = 1 - P(X < 3,6)$$~~

$$P(-1,2 < X < 3,6) = P\left(\frac{-1,2-7}{\sqrt{2}} < Z < \frac{3,6-7}{\sqrt{2}}\right) = P(-4,1 < Z < -1,7) = \Phi(-1,7) - \Phi(-4,1) = 0,4457 - 0 = 0,4457$$

EX2 $X \sim N(0, 1)$

$$P(X < -2,5) = \Phi(-2,5) = 0,00621$$

$$P(X > 4) = 1 - P(Z < 4) = 1 - \Phi(4) = 1 - 1 \approx 0$$

$$P(X > 0) = 1 - \Phi(0) = 1 - 0,5 = 0,5$$

$$P(X < 1,5) = \Phi(1,5) = 0,93319$$

$$P(X > -3,6) = 1 - \Phi(-3,6) = 1 - 0 \approx 1$$

$$P(-2,5 < X < 2,5) = \Phi(2,5) - \Phi(-2,5) = \Phi(2,5) - [1 - \Phi(2,5)] = 2\Phi(2,5) - 1 = 2 \times 0,99379 - 1$$

$$P(X \notin [-2, 2]) = 1 - P(-2 < X < 2) = 1 - [\Phi(2) - \Phi(-2)] = 1 - [2\Phi(2) - 1] = 2 - 2\Phi(2) = 2 - 2 \times 0,97725 =$$

EX3 $X \sim N(0, 1)$

$$P(X < n) = \Phi(n) = 0,896 \Rightarrow n = \Phi^{-1}(0,896) = 1,26$$

$$P(X > n) = 1 - P(X < n) = 0,1 \Rightarrow P(X < n) = 0,9 \Rightarrow n = \Phi^{-1}(0,9) = 0,67$$

$$P(-n < X < n) = 0,8 \Rightarrow 2\Phi(n) - 1 = 0,8 \Rightarrow 2\Phi(n) = 1,8 \Rightarrow \Phi(n) = 0,9 \Rightarrow n = \Phi^{-1}(0,9) = 0,67$$

$$P(X \notin [-n, n]) = 1 - P(-n < X < n) = 0,1 \Rightarrow 2\Phi(n) - 1 = 1 - 0,1 = 0,9 \Rightarrow 2\Phi(n) = 1,9 \Rightarrow \Phi(n) = 0,95 \Rightarrow n = \Phi^{-1}(0,95) = 1,65$$

EX4 $Y \sim N(20, 2)$

$$1) P(Y > 24) = 1 - P\left(Z < \frac{24-20}{\sqrt{2}}\right) = 1 - \Phi(2) = 1 - 0,97725$$

$$P(Y < 19) = P\left(Z < \frac{19-20}{\sqrt{2}}\right) = \Phi(-0,5) = 0,30854$$

$$P(19 < Y < 24) = P\left(\frac{19-20}{\sqrt{2}} < Z < \frac{24-20}{\sqrt{2}}\right) = \Phi(2) - \Phi(-0,5) = 0,97725 - 0,30854$$

$$1) P(20-a < X < 20+a) = 0,99$$

②

$$P\left(\frac{20-a-20}{\sigma} < Z < \frac{20+a-20}{\sigma}\right) = 0,99 \Leftrightarrow \Phi\left(\frac{a}{\sigma}\right) - \Phi\left(-\frac{a}{\sigma}\right) = 0,99$$

$$2\Phi\left(\frac{a}{\sigma}\right) - 1 = 0,99 \Rightarrow 2\Phi\left(\frac{a}{\sigma}\right) = 1,99 \Rightarrow \Phi\left(\frac{a}{\sigma}\right) = 0,995 \Rightarrow \frac{a}{\sigma} = \Phi^{-1}(0,995)$$

$$\frac{a}{\sigma} = 2,58 \Rightarrow a = 5,16$$

$$2) X \sim N(\mu, \sigma)$$

$$P(X < 15,84) = 0,1841 \Rightarrow P\left(Z < \frac{15,84 - \mu}{\sigma}\right) = 0,1841$$

$$P(X > 19,56) = 0,2578 \Rightarrow P\left(Z < \frac{19,56 - \mu}{\sigma}\right) = 0,7422$$

$$\begin{cases} \frac{15,84 - \mu}{\sigma} = \Phi^{-1}(0,1841) = -0,9 \\ \frac{19,56 - \mu}{\sigma} = \Phi^{-1}(0,7422) = 0,65 \end{cases} \Rightarrow \begin{cases} 15,84 - \mu = -0,9\sigma & (1) \\ 19,56 - \mu = 0,65\sigma & (2) \end{cases}$$

$$\Rightarrow \begin{cases} 3,72 = 1,55\sigma \Rightarrow \sigma = 2,4 & (2) - (1) \\ \text{replace } \sigma \text{ in } (2) \Rightarrow \mu = 18 \end{cases}$$

$$\text{Ex 5} \quad \begin{cases} P(X < 4,8) = 0,1 \\ P(X > 6,5) = 0,2 \end{cases} \Rightarrow \begin{cases} P\left(Z < \frac{4,8 - \mu}{\sigma}\right) = 0,1 \\ P\left(Z < \frac{6,5 - \mu}{\sigma}\right) = 0,8 \end{cases} \Rightarrow \begin{cases} \frac{4,8 - \mu}{\sigma} = \Phi^{-1}(0,1) \\ \frac{6,5 - \mu}{\sigma} = \Phi^{-1}(0,8) \end{cases}$$

$$\begin{cases} 4,8 - \mu = -1,28\sigma \\ 6,5 - \mu = 0,8\sigma \end{cases} \Rightarrow \begin{cases} 1,7 = 2,13\sigma \Rightarrow \sigma = 0,79 \\ \mu = 5,83 \end{cases}$$

$$\text{Ex 6} \quad X \sim N(\mu, \sigma) \quad \mu = 75 \quad C_V = 0,04 = \frac{\sigma}{\mu} \Rightarrow \sigma = 75 \times 0,04 = 3$$

$$X \sim N(75, 3)$$

$$a) \text{ Pieces Sales: } P(X < 72) = P\left(Z < \frac{72-75}{3}\right) = \Phi(-1) = 0,15866 = \frac{n_1}{1000} \Rightarrow n_1 = 158$$

$$\hookrightarrow \text{may } P(72 < X < 78) = P\left(\frac{72-75}{3} < Z < \frac{78-75}{3}\right) = \Phi(1) - \Phi(-1) = \frac{n_2}{1000}$$

$$0,84134 - 0,15866 = \frac{n_2}{1000} \Rightarrow n_2 = 682$$

$$P(X > 78) \Rightarrow n_3 = [1000 - (n_1 + n_2)] = 159$$

$$b) [75-k, 75+k]$$

$$P(75-k < X < 75+k) = 0,9 \Rightarrow P\left(\frac{75-k-75}{3} < Z < \frac{75+k-75}{3}\right)$$

$$P\left(-\frac{k}{3} < Z < \frac{k}{3}\right) = \Phi\left(\frac{k}{3}\right) - \Phi\left(-\frac{k}{3}\right) = 2\Phi\left(\frac{k}{3}\right) - 1 = 0,9 \Rightarrow 2\Phi\left(\frac{k}{3}\right) = 1,9$$

$$\Phi\left(\frac{k}{3}\right) = \frac{1,9}{2} = 0,95 \Rightarrow \frac{k}{3} = \Phi^{-1}(0,95) = 1,65 \Rightarrow k = 4,95$$

$$c) n=10 \quad p: \text{piece defectuosa} = 0,158 \leftarrow (\text{peu de pieces defectuoses})$$

$$X \sim B(10; 0,158)$$

$$P(X=4) = C_{10}^4 (0,158)^4 (0,842)^6$$

EX 7

$$X \sim N(5,35; 0,01)$$

$$\text{pieces defectives } \Phi[5,33; 5,37]$$

$$\frac{-0,02}{0,01} \quad \frac{+0,02}{0,01}$$

$$a) P(X \notin [5,33; 5,37]) \Rightarrow 1 - P(5,33 < X < 5,37)$$

$$= 1 - P\left(\frac{5,33-5,35}{0,01} < Z < \frac{5,37-5,35}{0,01}\right) = 1 - P(-2 < Z < 2)$$

$$= 1 - [\Phi(2) - \Phi(-2)] \Rightarrow 1 - [2\Phi(2) - 1] = 2 - 2\Phi(2) = 2 - 2 \times 0,97725$$

$$b) P(5,35-a < X < 5,35+a) = 0,8 \Rightarrow P\left(-\frac{a}{0,01} < Z < \frac{a}{0,01}\right) = 0,8$$

$$2\Phi\left(\frac{a}{0,01}\right) - 1 = 0,8 \Rightarrow \Phi\left(\frac{a}{0,01}\right) = \frac{1,8}{2} = 0,9 \Rightarrow \frac{a}{0,01} = \Phi^{-1}(0,9) = 1,29$$

$$a = 0,0129$$

EX 8 $X \sim N(0; 0,02)$

~~pieces defectives $\Phi[7,97; 8,03]$~~

~~$$1 - P(7,97 < X < 8,03) \Rightarrow 1 - P\left(\frac{7,97-0}{0,02} < Z < \frac{8,03-0}{0,02}\right)$$~~

~~$$= 1 - P(398,5 < Z < 401,5)$$~~

(4)

Ex 8 x : nombre d'heures en kg

$$X \sim N(500, 40)$$

$$1) P(X > 560) = 1 - P\left(Z < \frac{560 - 500}{40}\right) = 1 - \Phi(1,5) = 1 - 0,93319 = 0,06681 = \frac{n}{80}$$

$$\Rightarrow n \approx 5 \text{ veaux}$$

$$P(X < 480) = P\left(Z < \frac{480 - 500}{40}\right) = \Phi(-0,5) = 0,30854 = \frac{n}{80} \Rightarrow n \approx 25 \text{ veaux}$$

$$2) P(450 < X < 550) \Rightarrow P\left(\frac{450 - 500}{40} < Z < \frac{550 - 500}{40}\right)$$

$$= P(-1,25 < Z < 1,25) \Rightarrow 2\Phi(1,25) - 1 = 2 \times 0,89435 - 1 = 0,7887$$

$$3) P(X > n) = 0,15 \Rightarrow P(X < n) = 0,85$$

$$P\left(Z < \frac{n - 500}{40}\right) = 0,85 \Rightarrow \frac{n - 500}{40} = \Phi^{-1}(0,85) = 1,04$$

$$\Rightarrow n = 541,6 \text{ kg}$$

Ex 9

$$X \sim N(84, \sigma)$$

$$P(X < 64) = 0,16$$

$$P\left(Z < \frac{64 - 84}{\sigma}\right) = 0,16 \Rightarrow \Phi\left(-\frac{20}{\sigma}\right) = 0,16$$

$$-\frac{20}{\sigma} = \Phi^{-1}(0,16) = -1 \Rightarrow \sigma = 20$$

Ex 10

$$X$$
: nombre d'une follette $X \sim N(100, 1)$

$$1) P(98 < X < 102) = P\left(\frac{98 - 100}{1} < Z < \frac{102 - 100}{1}\right) = P(-2 < Z < 2) = \Phi(2) - \Phi(-2) = 2\Phi(2) - 1 = 2 \times 0,97725 - 1$$

$$2) P\left(\frac{98 - 100}{\sigma} < Z < \frac{102 - 100}{\sigma}\right) = 0,97 \Rightarrow P\left(-\frac{2}{\sigma} < Z < \frac{2}{\sigma}\right) = 0,97$$

$$2\Phi\left(\frac{2}{\sigma}\right) - 1 = 0,97 \Rightarrow \Phi\left(\frac{2}{\sigma}\right) = \frac{1,97}{2} = 0,985 \Rightarrow \frac{2}{\sigma} = \Phi^{-1}(0,985)$$

$$\frac{2}{\sigma} = 2,17 \Rightarrow \sigma = 0,92$$

EX 11

$$X \sim N(10; 0,06)$$

(5)

$$1) 1 - P(9,9 < X < 10,1)$$

$$1 - P\left(\frac{9,9 - 10}{0,06} < Z < \frac{10,1 - 10}{0,06}\right) = 1 - P(-1,66 < Z < 1,66)$$

$$\Rightarrow 2[1 - \Phi(1,66)] = 2 - 2 \times 0,95154 = 0,1$$

$$2) Y \sim N(10, \sigma)$$

$$3) Z = \frac{Y - 10}{\sigma} \sim N(0,1)$$

$$4) 1 - P(9,9 < X < 10,1) = 0,06$$

$$1 - P\left(\frac{9,9 - 10}{\sigma} < Z < \frac{10,1 - 10}{\sigma}\right) = 0,06$$

$$\Rightarrow P\left(-\frac{0,1}{\sigma} < Z < \frac{0,1}{\sigma}\right) = 0,94 \Rightarrow 2\Phi\left(\frac{0,1}{\sigma}\right) - 1 = 0,94$$

$$\Phi\left(\frac{0,1}{\sigma}\right) = \frac{1,94}{2} = 0,97 \Rightarrow \frac{0,1}{\sigma} = \Phi^{-1}(0,97) = 1,88$$

$$\sigma = 0,05$$